

REV2 REFERENCE / REDESIGN IN PROGRESS

22 September 2026: the current print order is N1 nose only.

Use Manufacturing/Nose_Print_N1/A59_Nose_N1_mm.stl and Nose_N1_Print_Guide.pdf. Quantity one, millimetres, 100%.

The following pages retain the old Rev2 design for reference. They are not the current full-aircraft manufacturing instructions. Their original page numbers remain unchanged; this notice is an additional page.

Superseded for the new design

Landing gear and steering system; the old 42-piece print order; A59 cover screw holes, columns, spacers and shaped washers; the large nose pitot-access notch. Do not fabricate these from the old instructions.

N1 restores the original upper nose shell. It uses a proposed simple ledge/tape attachment and a probe passage drilled to actual hardware during assembly. It retains the original rear mating envelope.

The rest of the aircraft will be simplified around the plywood skeleton and foamboard. Remaining mounts, control construction, belly protection, installation details and mass/CG estimates still require that redesign. Do not use the old mass, CG or gear clearance numbers for the new configuration.

Existing cutting templates are retained until a reuse/change list is complete. The N1 release is a nose-only handoff, not a completed redesign of every file in this manual.

Foam / form and fit

Stock

5 mm FliteBoard Pro paper-faced XPS, 1000 x600 mm. Measure thickness. Keep the outside paper as the developed reference; do not resize patterns for different stock.

Cut

Print at 100%, measure the 100 x20 mm rectangle and align matching registration marks. Knife-cut the CUT line. Mark the part ID and OUTSIDE before peeling.

Form

Test the tightest curve on scrap. Peel inside paper only where needed. Relieve/compress the core gradually over the real frame; do not stretch a short edge into place.

Seams

Bevel the inside core to the actual meeting angle. Butt outside-paper edges and add separate 15 mm internal paper straps. Keep adhesive and straps out of hinges. Hold alignment through cure.

Allowances

F23/F24 have a 1 mm paper-only lower lip. Other skins have no added lap/trim. F31/F32 are two-layer moving controls with 5 mm sacrificial stock before shaping. Use an XPS-compatible adhesive.

Foam / assembly sequence

Wings

Fit from the leading-edge region toward the open seam. Relieve only inner foam for ribs and hinge rails. On F01/F02, transfer A44 rib contacts to the inner core and relieve until dry-fit contact is removed; keep outside paper intact. Dry-fit a complete wing and moving surface before bonding. Check incidence, symmetry and twist.

Fuselage

Use the panel map. Fit lower forward and middle panels, then the rear underside. A42 underside order is F19, F20, F33, F21, F22. F27-F30 are the numbered rear-underside bands. Support every inner butt seam. Relieve only inside core in F19-F22/F33 and F27-F30 where A43 or A62R contacts it. Transfer these contacts from the assembled frame; do not cut the wood to force a skin fit.

Fixed tails

Use removable split collars on a straight jig: T03 at 6.84 mm, T02 at 171 mm, T01 at 335.16 mm along the 342 mm span. Remove both collar halves after cure. Keep the formed shape; do not add ribs into the filled section.

Moving tails

Laminate two 5 mm blanks per control. Trim to the finished outline and shape long edges to R5. Tape-hinge on the indicated axis, then fit backed horns and linkages. Verify combined travel.

Checks

Preserve outside paper around local reliefs. Reinforce seams without recreating a hard contact. Weigh completed skins and inspect them for warping or damage.

Foam / removable covers and ports

F25/F26

Join and form the two roof skins first. Make a service lid on the formed roof: X +/-25, Y=-115..-15. These are plan-view assembly coordinates, not a flat cutting pattern. Add 15 mm internal paper supports and replaceable reinforced-tape retention.

F23/F24

Join into one formed cover. Retain its perimeter on internal paper supports with six reinforced-tape tabs. Keep it removable for ESC/radio access. Do not permanently glue its lower lips to the body.

Cooling

On the formed rear cover, make a 24 x12 mm inlet at Y250 and a 30 x20 mm outlet at Y374. Add a small forward-facing scrap-foam scoop over the inlet, directing air toward the ESC. Confirm temperature in the loaded test.

Gear

Transfer actual installed gear crossings after forming. Give main legs 3 mm nominal clearance and the vertical nose shaft a 6 mm opening. Reinforce the paper border. Check steering and loaded deflection; skin must not carry landing loads.

Before closure

Prove battery insertion/removal, strap and stop access, sensor tubing, connectors, antenna support and full control travel. Leave pull strings and label all disconnects.